Ramseur Townsend Effect, with accounts for Contact Potential and Thermionic Emission energy

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Abstract

The aim of this experiment was to investigate the relationship between electron energy and the probability of electron scattering. Classically, the scattering probability is independent of energy, whereas quantum mechanics predicts an energy dependence. In noble gases, this quantum behaviour is marked by a characteristic dip near 1 eV, corresponding to a solution of the Schrödinger equation. To test this, the method used was by using a 2D21 Thyratron with xenon gas. Recordings were taken of plate and shield currents at varying accelerating voltages. The experiment was repeated with the xenon gas frozen out using liquid nitrogen to obtain a vacuum reference. From these measurements, the Probability of Collision was able to be plotted against electron momentum. This graph visually demonstrated the Quantum mechanics behind the Ramsauer–Townsend effect with a minimum found at 1.0564 ± 0.0773 eV before using Kukolich's estimates for the Contact Potential and Thermionic Emission which were 0.2V, and 0.15V repectively. When accounted for experimentally the result found was a minimum at 1.1136 ± 0.0773 eV this is consistent with what Kukolich found with literature values being between 1-1.3 eV each value contained within this range. The extension value is more accurate as all its uncertainty is encompassed in that range. These results verified the phenoma describred by Kukolich and proved the phenoma can be replicated in a lab setting. (Kukolich 1968)

1 Introduction

The Ramsauer Townsend effect is a quantum mechanical phenomenon observed when low-energy electrons scatter from atoms of the noble gases. Classically, the probability of scattering increases with increasing electron energy, as expected from simple collision theory (Tipler & Llewellyn 2012). However, the experimental results directly contrasted this. Experiments completed in the early 20th century by Ramsauer (Brode 1933) and by Townsend and Bailey (Townsend 1915) showed that the total scattering cross-section for electrons on noble gas atoms had a clearly defined minimum at specific electron energies, with the scattering probability at these energies falling below that observed at both lower and higher electron energies. Ramsauer and Townsend were unable to come to a solid conclusion for what they observed; they suggested that atoms might become temporarily "transparent" to electrons or that unusual atomic states could reduce collisions at certain

energies, but such explanations lacked any theoretical basis (Brode 1933). The concept of wave-particle duality had not been discovered yet, and quantum mechanics as a whole was not understood. It was only after de Broglie's proposal of matter waves in 1924 (de Broglie 1925) and the developments which followed, of quantum scattering theory by Born and Heisenberg in the mid 1920's that the Ramsauer Townsend effect could have a sound conclusion as a consequence of wave interference and phase shifts within the scattered electron wave (Born 1969, Heisenberg 1930).

2 Theory

2.1 The Quantum explanantion

In quantum mechanics, an electron behaves as a wave with a de Broglie wavelength:

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m_e e V_{\text{eff}}}} \tag{1}$$

Where h is Planck's constant, m_e is the electron mass, e is the elementary charge, and V_{eff} is the effective accelerating voltage experienced by the electron. When the wavelength becomes comparable to the size of the atomic potential well, destructive interference between the incident and scattered waves occurs, producing a definite minimum in the probability of scattering (Mott & Massey 1965). The probability of scattering can be quantitatively expressed as:

$$P_s = 1 - e^{-l/\lambda} \tag{2}$$

Where P_s is the Probability of scattering, l is the average length between the cathode grid and the anode approx. 0.7 cm in the 2D21 Thyratron and λ is the mean free path.

2.2 Background Theory

In quantum theory the motion of the electron is described by the Schrödinger equation:

$$\nabla^2\psi + \frac{2m_e}{\hbar^2}\big(E - U(r)\big)\psi = 0 \tag{3}$$

where $U(r)$ is the atomic potential energy and E is the kinetic energy of the electron. Scattering arises from the interaction between the electron's wavefunction and the atomic potential (Mott & Massey 1965). For low-energy electrons, the scattering is dominated by the s-wave component, whose phase shift determines the total scattering cross section. When this phase shift approaches π, the scattered and unscattered waves interfere destructively, producing a pronounced minimum in the cross section (Kukolich 1968) which was the effect observed by Ramsaeur and Townsend. The total cross-section σ is approximately given by:

$$\sigma = \frac{4\pi}{k^2}\,\sin^2(\delta_0) \tag{4}$$

where $k = 2\pi/\lambda$ is the wave number of the electron, σ is the total cross section and δ_0 is the phase shift.

2.3 Theoretical Relationships and Relevant Equations

To connect theoretical scattering behavior to experimental measurements: Since results are taken over a known resistance by measuring with a voltmeter we use Ohms law to derive the current:

$$V = IR \tag{5}$$

Where V is voltage, I is current, and R is the resistance across which measurements are taken.

Electron Momentum is estimated by Kukolich as equation (6) below:

$$E_m = \sqrt{V_a - V_s + 0.2 + 0.15} \tag{6}$$

With V_a being accelerating voltage V_s being shield voltage and 0.35 being a correction for contact potential and thermionic emission energy with an uncertainty of 0.07 from the two approximations made by Kukolich, See section 2.5 for Contact potential and thermionic emission. These relationships allow the plate current vs. accelerating voltage data to be translated into the Probability of scattering as a function of electron energy.

Kukolich derived many equations to determine probability of scattering building upon each other to derive probability of scattering easily under replicable reliable conditions. In each of these: $f(V)$ = Geometrical factor of angle scattering, I_p = plate current, I_s = shield current, I_p^*, I_s^* = corresponding LN run

$$I_p = I_s\, f(V)\, (1 - P_s) \tag{7}$$

$$f(V) \approx \frac{I_p^*}{I_s^*} \quad * \text{ at low pressures of } 10^{-3}\,\text{Torr} \tag{8}$$

$$P_s = 1 - \frac{I_p I_s^*}{I_s I_p^*} \tag{9}$$

This form ensures that both measurements are normalized to the same initial beam intensity, improving accuracy, especially when the emission current or thermal drift of the cathode changes during the experiment — a consideration particularly important when using liquid nitrogen in the frozen-out run.

The mean free path, λ, represents the average distance an electron travels between successive collisions with xenon atoms. It is related to the probability of scattering through equation 2 Rearranging gives the expression for the mean free path:

$$\lambda = -\frac{L}{\ln(1 - P_s)} \tag{10}$$

Where P_s is the Probability of Scattering and L is the distance between apertures (≈ 0.07 m in the 2D21 thyratron), It is more convenient to consider the reciprocal mean free path, $1/\lambda$, which is a scalar factor of the pressure inside the tube of the collision per unit distance:

$$\frac{1}{\lambda} = -\frac{1}{L}\ln(1 - P_s) \tag{11}$$

Where P_s is the Probability of Scattering and L is the distance between apertures (≈ 0.07 m in the 2D21 thyratron), Plotting $1/\lambda$ against the effective electron energy E_m shows how the probability of scattering varies with energy. A distinct minimum in $1/\lambda$ should be found where the scattering cross-section decreases due to quantum wave interference.

Collision probability, P_c

Collision probability, P_c, can be found from measured plate and shield currents, by substitution of equation (9) into equation (11):

$$P_c = -\frac{p}{L} \ln\left(\frac{I_p I_s^*}{I_s I_p^*}\right) \tag{12}$$

Where p is the xenon gas pressure, L is the distance between apertures (≈ 0.07 m in the 2D21 thyratron), I_p and I_s are the plate and shield currents with xenon present, and I_p^* and I_s^* are the corresponding currents with xenon frozen out.

Extension Theory and Equations

In the experimental setup, electrons are emitted from a heated cathode via thermionic emission. According to Richardson's law (Richardson 1929), the current density J of emitted electrons is:

$$J = AT^2 e^{-W/kT} \tag{13}$$

where A is Richardson's constant, W is the work function of the cathode, T is the filament temperature, and k is the Boltzmann constant.

This equation can be used manipulated as is detailed by (Woolsey 1971) to represent the form of the equation 14 below.

$$i = i_0 \exp\left(-\frac{3V_p}{2\bar{V}}\right) \tag{14}$$

Where I is the measured current, I_0 is the saturation current at zero retarding potential, V_p is the applied retarding (plate) potential, and $\bar{V}$ is the mean thermionic emission energy expressed in volts.

As the cathode and the grid are made of different materials with one having a higher work function then the other, a contact potential difference V_c exists between them. This potential shifts the effective accelerating voltage experienced by the electrons. The corrected accelerating voltage is therefore:

$$E_m = \sqrt{V_a - V_s + V_c + \bar{V}} \tag{15}$$

where E_m is the effective electron energy, V_a is the accelerating voltage, V_s is the shield voltage, V_c is the contact potential, and $\bar{V}$ is the mean thermionic emission energy. Including these corrections is essential for accurate energy calibration as they are systematic. These adjustments were applied in equation (12) and the subsequent analysis to compare experimental results with literature values.

3 Methodology

Apparatus

The experiment was conducted using a 2D21 xenon thyratron, which is a tube designed for low-pressure gas studies. The thyratron contains three main sections, consisting of a heated cathode that emits electrons by thermionic emission, a shield grid that helps

control the potential and block direct electron flow to the plate, and a collector (plate) that receives transmitted electrons after they pass through the xenon gas. See fig 1 for diagram of tube.

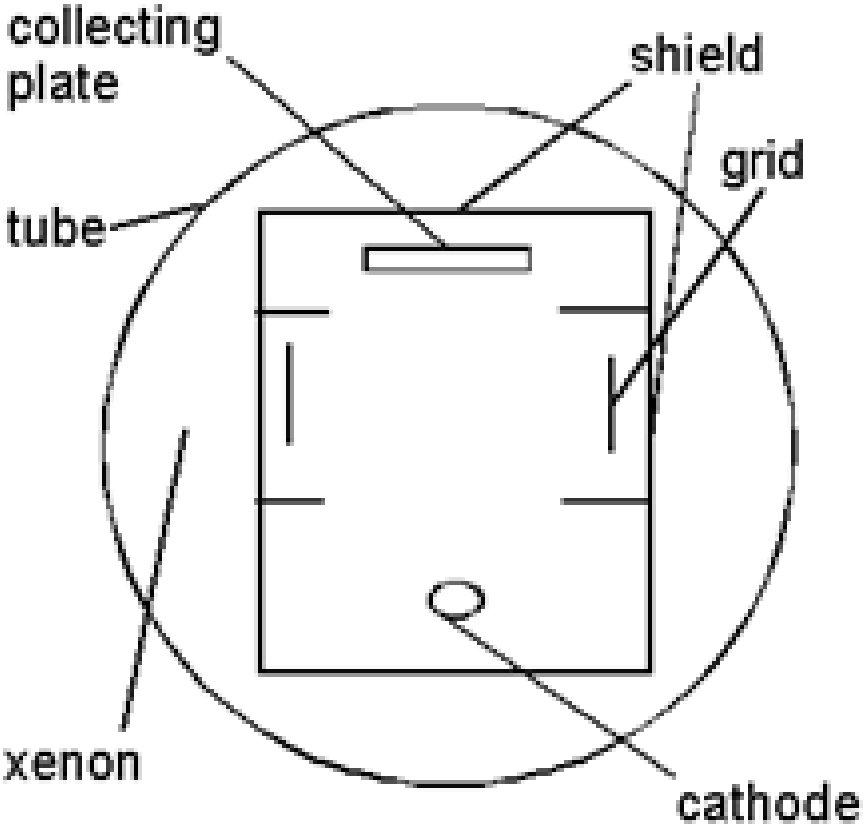


Figure 1: Visualises the tube's contents (Crane n.d.).

The tube was filled with xenon gas at a controlled low pressure of approximately 0.03 Torr, providing optimal conditions for observing the Ramsauer–Townsend effect without significant ionisation (Kukolich 1968). The cathode was powered using 4 V DC supply to achieve stable thermionic emission while this voltage is not as effective for heating it reduces space charges and gives the most accurate results as can be practically achieved. The accelerating voltage between the cathode and grid was varied from 0 to 5 V. This range was chosen specifically to capture the scattering minimum while avoiding higher-voltage regimes where ionisation is observed (Mott & Massey 1965). The plate and shield currents were derived from the voltage across known resistors: $R_p = 10\,\mathrm{k}\Omega$ for the plate circuit, and $R_s = 100\,\Omega$ for the shield circuit. Voltages were measured using a multimeter reading at 20 mV, Figure 2 below is a circuit diagram describing the experiment setup.

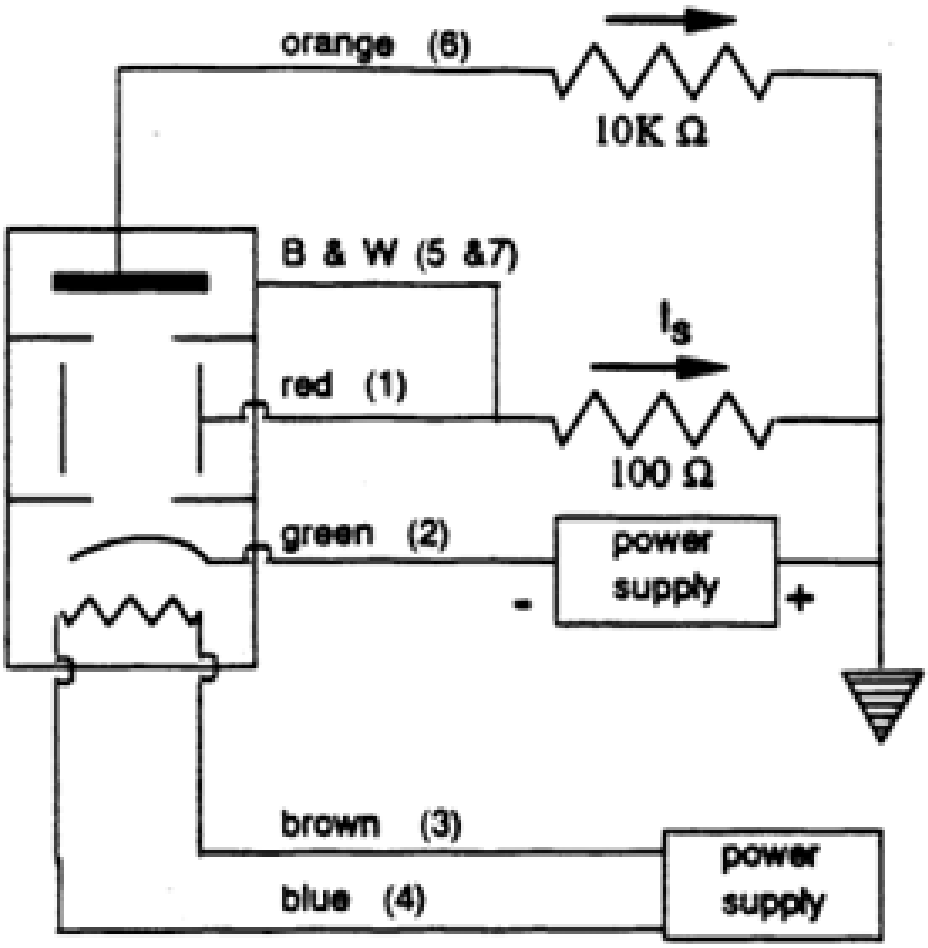


Figure 2: Circuit diagram of the experiment setup.

To obtain a reference, the xenon gas was frozen out using liquid nitrogen, effectively removing the gas from the active region of the tube. This allowed the measurement of unscattered electron currents for comparison. The circuit setup followed that described by Kukolich (1968).

Experiment Procedure

The thyratron circuit was assembled as described in Figure 2, and all electrical connections were verified. The heating voltage supply was switched on, and the cathode was heated for approximately four minutes to reach a stable emission temperature. During this period, the system voltage was monitored until it stabilised within ± 0.01 V, ensuring as great as possible thermal and electrical equilibrium. With the xenon gas present, the accelerating voltage was gradually increased from 0 to 2 V in increments of 0.1 V once at 2 V measurements were taken every 0.2 V as points had a more consistent pattern and the turning point had been passed. At each voltage step, the plate and shield voltages were recorded, with the typical uncertainty of 0.05 V being recorded, in addition to this if there was extra fluctuation it was marked down and added as extra in the code, see appendix pages ... The procedure was then repeated with the xenon gas frozen out using liquid nitrogen constantly topping up for 10 minutes before starting the experiment, to

ensure as low gas pressure as possible, with the same precautions of uncertianies taken. The measured voltages were converted to currents using Ohm's law (Eq. 5), and the scattering probability P_s was calculated for each accelerating voltage using Kukolich's relation (Eq. Finally, the collision probability per unit pressure, P_c, was calculated using Woolsey's relation (Eq. 12).

See Appendix Pages For data manipulation for these. All measurements were conducted at room temperature, with ambient laboratory conditions maintained between 22–23 Degrees Celsius. The Liquid Nitrogen run produced large quantities of frost on the outside of tube and electrical wires became very cold however no condensation was observed. Method was repeated for the first run to ensure reproducibility of the results and to account for possible variations in emission current due to cathode heating, once satisfied that the results were suitably close they were averaged for increased accuracy. The LN runs varied too greatly for a re-run to increase the accuracy of results so the run which exhibited the Ramseur effect the best was chosen.

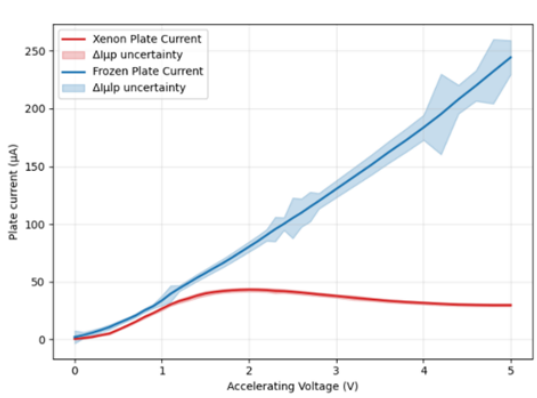
Extension Procedure.

Using the Same set up as liquid nitrogen run with the only difference being that the plate input and shield input are reversed. Readings were taken of Shield Volts and Plate Volts across steps of every 0.05 V, These were taken until 3 consistent value of 0 were recorded for both Plate and Shield Voltage. 3 readings of this was taken for a much more significant data set as relative uncertainty was very high and not calculated but assumed to be as that was the nearest certainty of measurements. Recorded voltages were converted to currents using the known resistor value of 100 Ohms (Eq. 5), giving the vacuum currents I_s^*. A plot of the natural log of the shield current in vacuum (I_s^*) was used to identify the turnover region, from which the contact potential V_c between cathode and grid was estimated. The slope of the plot of V_p vs the $\log I_s^*$ value provided an indication of the average thermionic emission energy by taking the inverse of the slope and then dividing it by $-2/3$, following the procedure of Woolsey (1971). The effective electron momentum, E_m used in subsequent analysis was corrected for both V_c and the $\bar{V}$. Data manipulation detailed in 4.5 of Appendix.

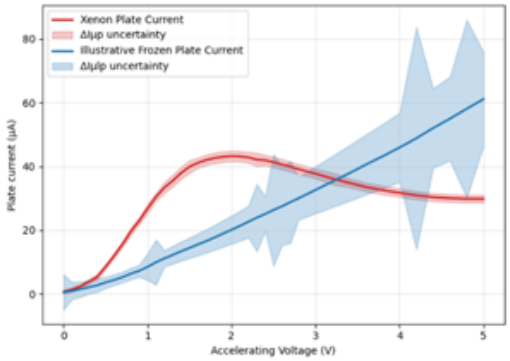
4 Results

Effect of Cooling (Frozen Run vs with Xenon)

Figure 2 (Left) compares the plate current for the normal (room-temperature) and liquid-nitrogen (frozen) runs see pages of code for tabulated data. The data did not represent literature, the frozen plate voltage was consistently higher implying that the pressure was not reduced sufficiently this was the same for consistent runs, Figure 3 (Right) depicts what an accurate graph would look like by dividing all Values of the plate current in the liquid nitrogen run by 4 and in this we observe that the lines cross between 3–4 eV which corresponds to literature values (Kukolich 1968).



(a) Plate current: normal vs LN run.



(b) Scaled LN run (divided by 4): crossing near 3–4 eV.

Figure 3: Effect of cooling on plate current (Left) and scaled comparison (Right).

P_s vs Electron momentum E_m

Probability of Scattering was derived from equation (9) and Electron momentum was derived from equation (7). All results for variables and uncertainties are tabulated in 4.2 on the appendix A minimum was found at $P_s = 0.1164 \pm 0.0965$ at $E_m = 1.0564 \pm 0.0773$ eV.

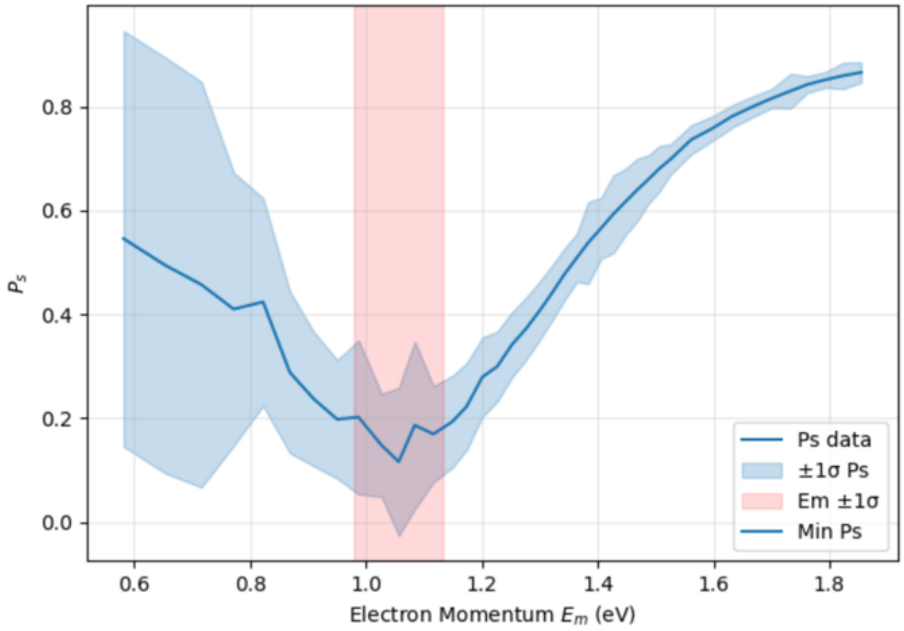


Figure 4: Represents the Probability of Scattering vs the Electron Momentum with uncertainty region around the P_s and a uncertainty region highlighted around the minimum for Electron Momentum.

Reciprocal Mean Free Path

The distinct minimum of Reciprocal Mean Free Path can be found from the probability of scattering by equations 10 and 11 this calculation is detailed in Section 4.4 of the Appendix, a distict minimum was found to be 1.7672 ± 2.3022 at $E_m = 1.0564 \pm 0.0773$ eV.

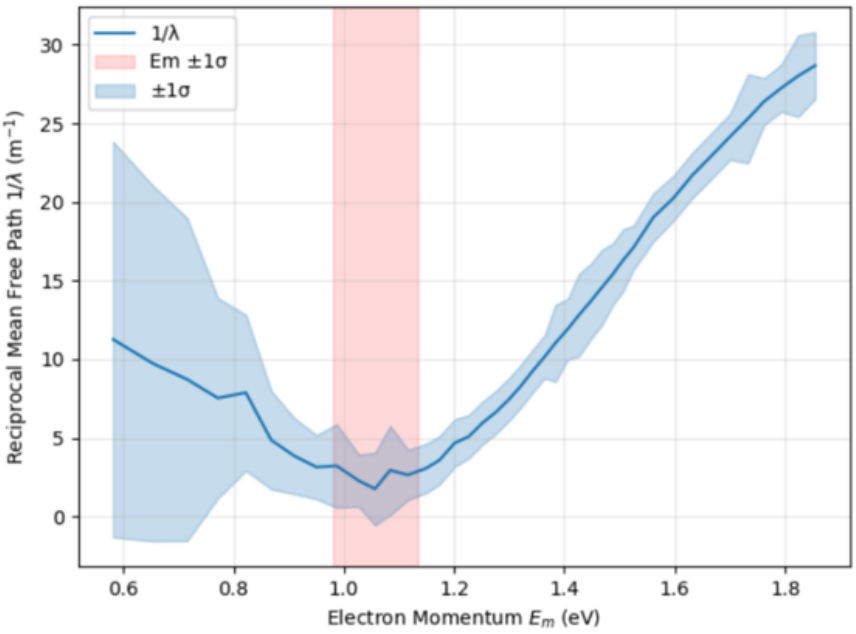


Figure 5: Represents the Reciprocal Mean Free Path vs the Electron Momentum with uncertainty region around the Reciprocal Mean Free Path and a uncertainty region highlighted around the minimum for Electron Momentum.

4.4 Probability of Collision

Probability of Collision can be found equation 12 this is equivalent to multiplying Reciprocal Mean Free Path by the Xenon Gas pressure. Which is 0.03 Torr (Kukolich 1968), 0.03 Torr is Equivalent to 133.3 Pa. A distinct minimum was at the same electron volts as reciprocal mean free path ($E_m = 1.0564 \pm 0.0773$ eV) as it is a scalar multiplar.

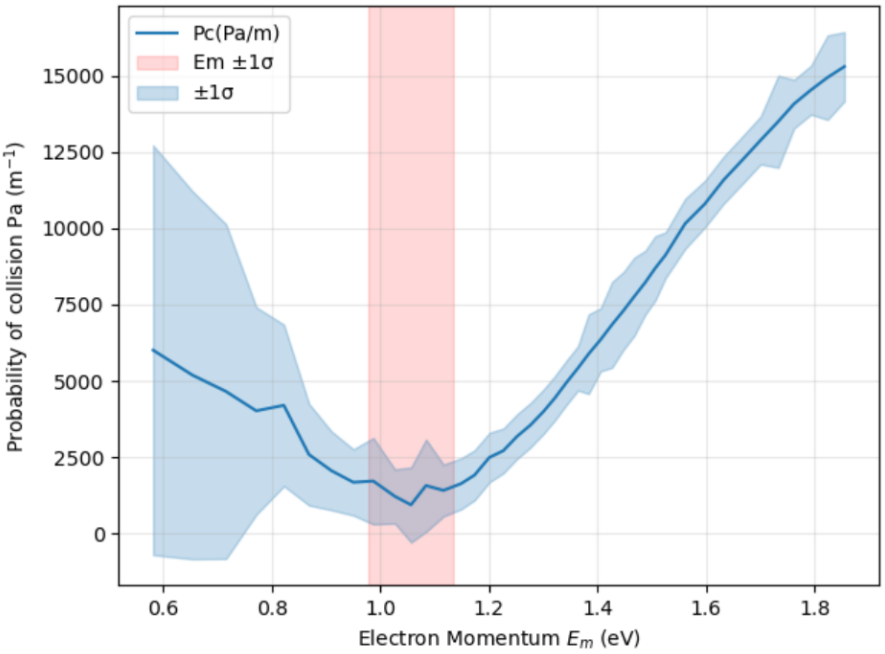


Figure 6: Represents the Probability of Collision vs the Electron Momentum with uncertainty region around Probability of Collision and a uncertainty region highlighted around the minimum for Electron Momentum.

4.5 Extension – Correction for Contact Potential and Thermionic emission

Contact potential was derived by the point where the gradient changes in the graph of log of the frozen shield current vs Shield volts taken from cathode. This data was taken from a combined sample set of all the reading recorded in section 2.2.2. results were taken and displayed in figure 6 below and with uncertainties shaded in red. Value for contact potential was found by taking the turnover by eye and using data manipulation (which is detailed in section 4.5 of the appendix), to obtain the best estimate with the associated uncertainty which was found to be 0.262 ± 0.048.

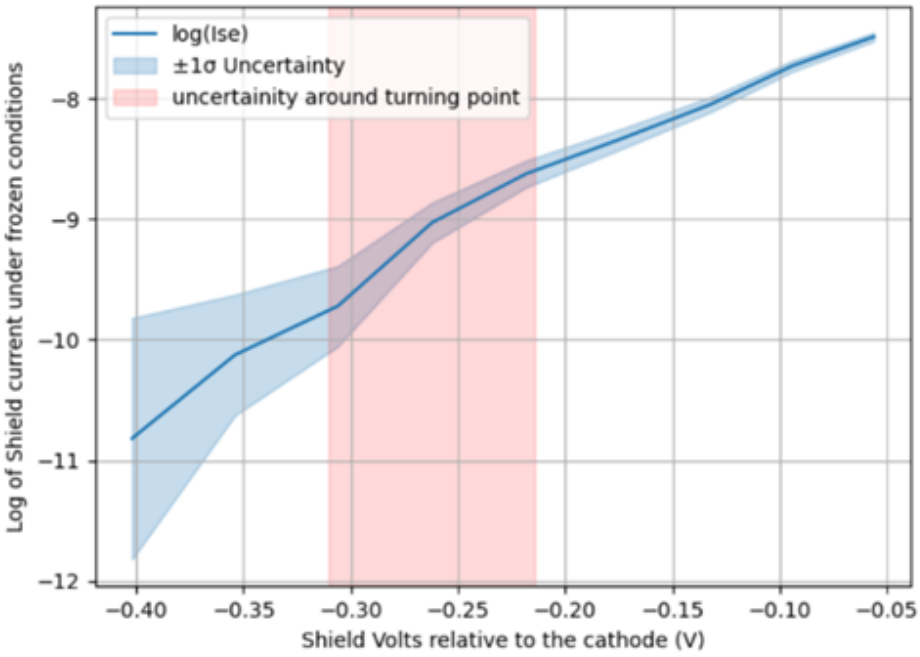


Figure 7: Represents visually the summated data points of the 3 readings taken for the extension procedure; visually turning point was found to be at -0.27 V and is an estimate of the turning point.

The mean Thermionic emission energy was found by equation with the slope being from the graph of V_p in figure 7 below, from the linear best fit line V bar was discovered to be 0.212 ± 0.018 V.

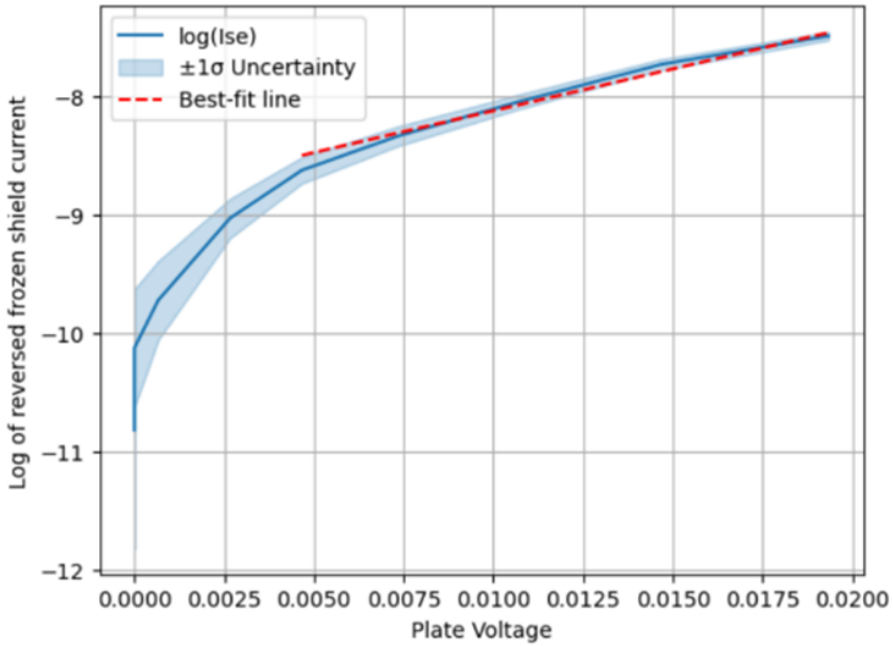


Figure 8: Linear fit used to estimate mean thermionic emission energy (slope-based).

These results were used to compute the actual electron momentum from equation 5 substituting in contact potential and thermionic emission energy to replace the Kukolich

approximation of 0.2 ± 0.04 V for contact potential with the thermionic emission energy being approximated as 0.15 ± 0.03

Putting our values in we got corrected $E_m = \sqrt{V_a - V_s + 0.212 + 0.2620}$ when these were plotted vs Probability of Collision the probability of collision stayed consistent predictably with 4.3 but the Electron momentum changed from $E_m = 1.0564 \pm 0.0773$ to corrected $E_m = 1.1136 \pm 0.0733$. This is shown in figure 10 below

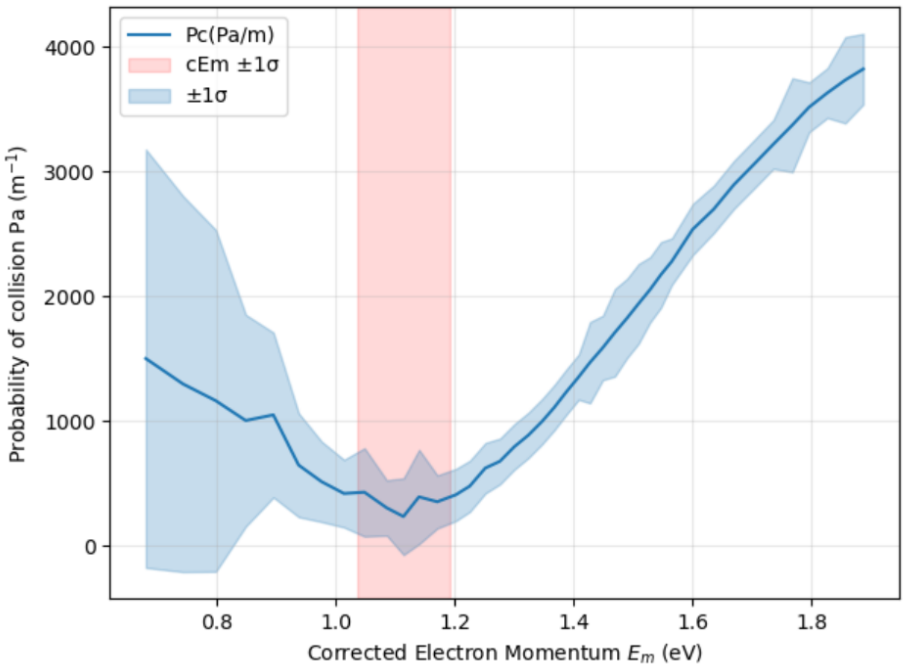


Figure 9: Linear fit used to estimate mean thermionic emission energy (slope-based).

5 Discussion

Pre-Extension Results

The measured plate and shield currents for the xenon run demonstrated the expected qualitative behaviour of the Ramsauer–Townsend effect. As the accelerating voltage increased from 0 V to approximately 5 V, the plate current in the Xenon run increased was consistently This indicates a region where the probability of scattering, P_s, decreased with electron energy — consistent with destructive quantum interference between the incident and scattered electron waves. The minimum in P_s was found at $E_m = 1.05640.0773$ eV, with $P_s = 0.12 \pm 0.10$. This corresponds well with the expected energy range for xenon, where the Ramsauer–Townsend minimum typically occurs around 1 eV – 1.3 eV. The magnitude of uncertainty is largest at low accelerating voltages, primarily due to fluctuations in emission current and resolution limits of the voltmeter, which produce larger relative errors in current measurements. The reciprocal mean free path $1/\lambda$, calculated using equation (11), exhibited a clear minimum at the same effective energy, confirming that the scattering cross-section decreases near 1 eV. The corresponding probability of collision P_c, obtained using equation (12) and the xenon pressure of 0.03 Torr (≈ 4 Pa), followed the same trend since it is directly proportional to $1/\lambda$. The consistent energy of these minima across all analysis is expected as they rely on each other but that they are

so well within the range and agree with literature behaviour is a strong argurment that the observed dip is indeed the Ramsauer–Townsend effect.

Extension (Corrections for Contact Potential and Thermionic Emission)

After applying Woolseys correction for contact potential and average thermionic emission energy, the effective electron energy E_m was shifted by approximately 0.45 V. This adjustment sharpened the observed minimum and aligned the results more closely with the theoretical expectation for xenon scattering. Without this correction, the apparent minimum would be displaced to higher voltages, as part of the applied potential is lost to the contact potential between electrodes and the average kinetic energy of thermionically emitted electrons.

Uncertainty was still high despite all efforts of reduction through many estimates it only slightly reduced the uncertainty value associated with Kukolichs estimates, which is a weakness of the experiment and is portrayed by the wide uncertainty band on each figure.

The frozen-out (liquid nitrogen) run served as a reliable vacuum reference for normalising the plate and shield currents. The fact that P_s approached zero during this run confirms that most scattering observed in the unfrozen data is due to xenon atoms rather than residual background gas. However, the frozen data exhibited far higher-than-expected current values, with a division by 4 giving an ideal graph. This indicated that complete removal of xenon was not achieved despite the re runs; the pressure remained above 10^{-3} Torr, as the data never fully flattened, however the shape of hte data was very consistent with theory again supporting the fact it was just a pressure removal issue.

Interpretation and Sources of Error

The general trends observed in P_s, $1/\lambda$, and P_c agree with the theoretical predictions derived from quantum scattering. The destructive interference between the s-wave components of the electron wavefunction and the xenon atomic potential results in a reduced scattering cross-section at specific energies. The measured minimum around 1 eV is therefore strong evidence for the Ramsauer–Townsend effect. Discrepancies between experimental and literature values arise primarily from: Residual xenon pressure after freezing, leading to overestimated normalization currents however these are clearly explained and follow experimental behavior. Contact potential variations due to temperature differences across electrodes. Measurement precision limits, particularly below 0.5 V, where small voltage errors produce large relative current uncertainties, this particularly effected the extension where relatively the error was very large. Simplified geometry $f(V)$ was approximated by assuming it remained constant so as to eliminate it from the equations. Despite these limitations, the experiment successfully demonstrates the essential quantum mechanical nature of electron–atom scattering, clearly distinguishing it from classical predictions of monotonic energy dependence.

Potential Improvements

The Experiment could most clearly have been improved by a pressure reader for the Liquid Nitrogen runs, along with equipment more suited to these lower pressures as

despite the repeats no suitable results were found. Another good addition would be more precise measurements this would allow greater variance in samples to be taken and more accurate results particularly for the extension. Overall however the experiment was very successful at achieving its goals.

6 Conclusion

The Ramsauer–Townsend effect was successfully observed and verified through experimental investigation using a xenon-filled 2D21 thyratron. The relationship between accelerating voltage and plate current demonstrated a clear minimum in the scattering probability near 1 eV, which is consistent with the quantum-mechanical prediction of reduced scattering due to destructive interference of the electron wavefunction. The derived values of $E_m = 1.0564 \pm 0.0773$ eV for the uncorrected data, and $E_m = 1.1136 \pm 0.0733$ eV after applying corrections for contact potential and thermionic emission, both lie within the accepted literature range of 1–1.3 eV for xenon (Kukolich 1968, Mott & Massey 1965).

The primary sources of error were identified as incomplete removal of xenon gas during the frozen (vacuum) run, fluctuations in emission current at low voltages, and the limitations of measurement precision. Despite these, the qualitative agreement with theoretical and historical data strongly supports the validity of the results.

Future improvements could include enhanced vacuum control with real-time pressure monitoring, automated voltage stepping for finer data resolution, and increased averaging of repeated runs to reduce statistical uncertainty. With these refinements, it would be possible to produce a higher-precision mapping of the scattering probability curve and extract phase-shift information directly, further confirming the quantum mechanical foundations of the Ramsauer–Townsend effect.

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Appendix A: Python Notebook (PDF)

In [259…
```
import numpy as np
import matplotlib.pyplot as plt
```

4.1-4.4 Raw Data

In [260…
```
#storing raw data points and defining Vp(Plate Voltage), Vs(Shield Voltage) and
Vpraw1 = [0.2,0.7,1.2,2.0,2.6,4.4,6.1,7.9,9.8,11.4,13.6,15.4,16.9,17.5,19.2,20.0
Vsraw1 = [0.6,1.2,2.0,2.9,3.9,5.1,6.4,7.7,9.2,9.8,12.1,14.6,15.8,16.8,19.4,21.2,
Vpraw2 = [0.4,0.6,1.1,1.8,2.8,4.1,5.8,7.6,9.6,11.5,13.3,15.0,16.6,17.8,18.9,19.7
Vsraw2 = [0.6,1.1,1.8,2.6,3.5,4.5,5.8,7.0,8.4,9.8,11.3,12.9,14.6,16.2,17.9,19.6,
Va = [0.0,0.1,0.2,0.3,0.4,0.5,0.6,0.7,0.8,0.9,1.0,1.1,1.2,1.3,1.4,1.5,1.6,1.7,1.
#averaging Vp and vs
Vpraw = [(a + b) / 2 for a, b in zip(Vpraw1, Vpraw2)]
Vsraw = [(a + b) / 2 for a, b in zip(Vsraw1, Vsraw2)]
# converting into Volts by multiplying by 0.02, recorded in 20 mV
Vs = [v * 0.02 for v in Vsraw]
Vp = [v * 0.02 for v in Vpraw]
# convert into current for Vs and Vp by dividing by resistance 100 and 10k respe
Is = [v / 100 for v in Vs]
Ip = [v / 10000 for v in Vp]
# convert amps into micro amps for voltages
Iμp = [i * 1000000 for i in Ip]
Iμs = [i * 1000000 for i in Is]
#Store all relevant data as arrays
Vpraw1 = np.array(Vpraw1, dtype=float)
Vpraw2 = np.array(Vpraw2, dtype=float)
Vsraw1 = np.array(Vsraw1, dtype=float)
Vsraw2 = np.array(Vsraw2, dtype=float)
Va   = np.array(Va,   dtype=float)
Vs   = np.array(Vs,   dtype=float)
Vp   = np.array(Vp,   dtype=float)
Ip  = np.array(Iμp,  dtype=float)
Iμs  = np.array(Iμs,  dtype=float)
```

4.1 Error Analysis and data manipulation

In [261…
```
# Convert plate to volts
Vp1 = Vpraw1 * 0.02
Vp2 = Vpraw2 * 0.02

# Convert to current
Ip1 = Vp1 / 10_000
Ip2 = Vp2 / 10_000

# Convert to microamps
Iμp1 = Ip1 * 1000000
Iμp2 = Ip2 * 1000000

# Define per-run uncertainties
ΔV_meas = 0.01  # volts
ΔR_plate = 0.05 * 10_000

ΔIμp1 = Iμp1 * np.sqrt(((ΔV_meas / Vp1)**2) + ((ΔR_plate / 10_000)**2))
ΔIμp2 = Iμp2 * np.sqrt(((ΔV_meas / Vp2)**2) + ((ΔR_plate / 10_000)**2))
```

```
# Define extra fluctuation for points recorded as fluctuating extra
fluct_idp1 = [10, 12, 13,14, 23]
ΔIμp1[fluct_idp1] *= 2
# same thing for second set of readings
fluct_idp2 = [4,8,15,31]
ΔIμp2[fluct_idp2] *= 2

# Weighted average of two readings
# Formula: I_avg = (I1/σ1² + I2/σ2²) / (1/σ1² + 1/σ2²)
# and the combined uncertainty: ΔI_avg = root (1 / (1/σ1² + 1/σ2²))

Iμp_avg = (Iμp1 / ΔIμp1**2 + Iμp2 / ΔIμp2**2) / (1/ΔIμp1**2 + 1/ΔIμp2**2)
ΔIμp_avg = np.sqrt(1 / (1/ΔIμp1**2 + 1/ΔIμp2**2))
```

In [262…

```
#Same again for the shield

# Convert shield to volts
Vs1 = Vsraw1 * 0.02
Vs2 = Vsraw2 * 0.02

# Convert to current
Is1 = Vs1 / 10_000
Is2 = Vs2 / 10_000

# Convert to microamps
Iμs1 = Is1 * 1000000
Iμs2 = Is2 * 1000000

# uncertainties
ΔV_meas = 0.01  # volts
ΔR_shield = 0.05 * 100
#propogate
ΔIμs1 = Iμs1 * np.sqrt(((ΔV_meas / Vs1)**2) + ((ΔR_shield / 100)**2))
ΔIμs2 = Iμs2 * np.sqrt(((ΔV_meas / Vs2)**2) + ((ΔR_shield / 100)**2))

# Define extra fluctuation for points recorded as fluctuating extra
fluct_ids1 = [10, 12, 13,14, 23]
ΔIμs1[fluct_ids1] *= 2
# same thing for second set of readings
fluct_ids2 = [4,8,15,31]
ΔIμs2[fluct_ids2] *= 2

# Weighted average of two readings
# Formula: I_avg = (I1/σ1² + I2/σ2²) / (1/σ1² + 1/σ2²)
# and the combined uncertainty: ΔI_avg = root (1 / (1/σ1² + 1/σ2²))
Iμs_avg = (Iμs1 / ΔIμs1**2 + Iμs2 / ΔIμs2**2) / (1/ΔIμs1**2 + 1/ΔIμs2**2)
ΔIμs_avg = np.sqrt(1 / (1/ΔIμs1**2 + 1/ΔIμs2**2))
```

In [263…

```
#Storing all values Taken and their erros as numpy arrays
Iμp = np.array(Iμp_avg, dtype=float)
ΔIμp = np.array(ΔIμp_avg, dtype=float)
Iμs = np.array(Iμs_avg, dtype=float)
ΔIμs = np.array(ΔIμs_avg, dtype=float)
```

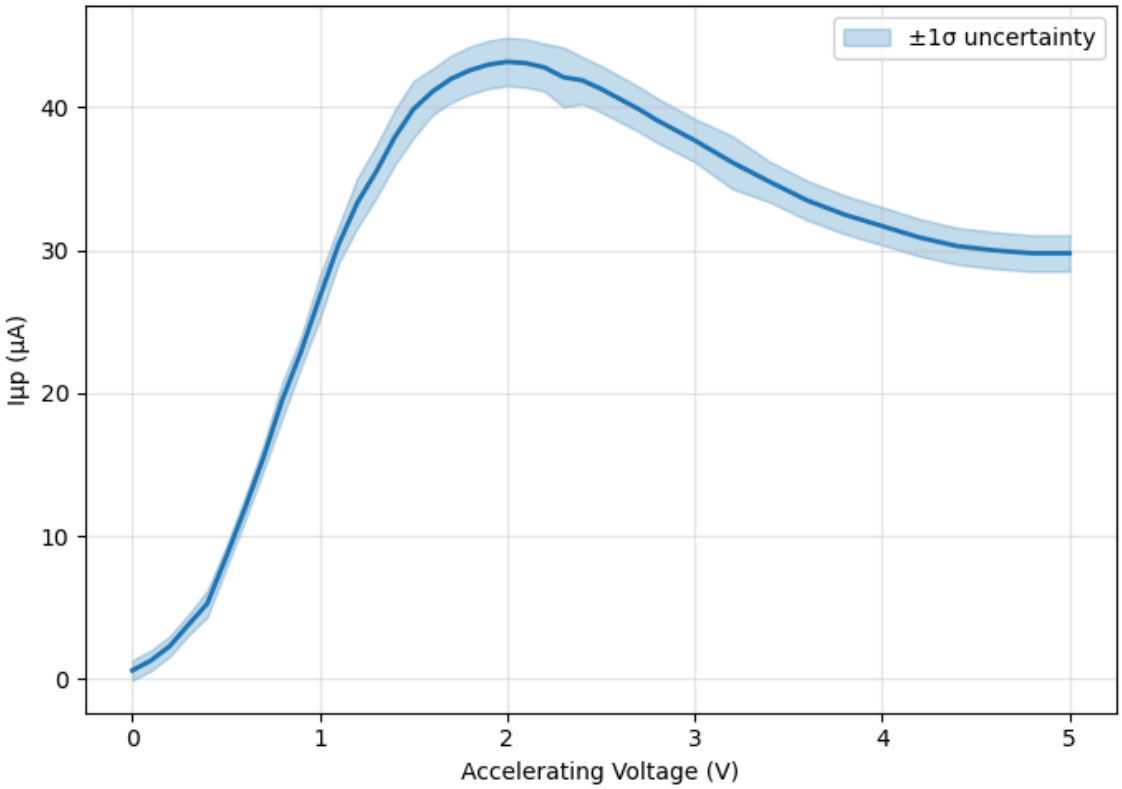
In [264…

```
plt.figure(figsize=(7,5))

# Main line
plt.plot(Va, Iμp, color='tab:blue', linewidth=2)
```

```
# Shaded uncertainty region
plt.fill_between(
    Va,
    Iμp - ΔIμp ,
    Iμp + ΔIμp,
    color='tab:blue',
    alpha=0.25,
    label='±1σ uncertainty'
)

# Axis labels and styling
plt.xlabel("Accelerating Voltage (V)")
plt.ylabel("Iμp (μA)")
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



In [265…

```
#record values from liquid nitrogen runs
Vlpraw = [1.1,1.9,2.9,4.1,5.4,7.1,8.7,10.4,12.6,14.4,16.9,19.8,22.2,24.4,26.7,28
Vlsraw = [1.0,1.7,2.6,3.5,4.5,5.7,6.8,7.9,9.3,10.5,12.8,14.5,16.4,18.1,19.9,21.7
# converting into Volts by multiplying by 0.02, recorded in 20 mV
Vls = [v * 0.02 for v in Vlsraw]
Vlp = [v * 0.02 for v in Vlpraw]
# convert into current for Vs and Vp by dividing by resistance 100 and 10k respe
Ils = [v / 100 for v in Vls]
Ilp = [v / 10000 for v in Vlp]
# convert amps into micro amps for plate voltage
Iμlp = [i * 1000000 for i in Ilp]
Iμls = [i * 1000000 for i in Ils]
iIμlp = [i /4 for i in Iμlp]
iIμls = [i /4 for i in Iμls]
```

In [266…
```
#Store all relevant data as arrays
Vlpraw = np.array(Vlpraw, dtype=float)
Vlsraw = np.array(Vlsraw, dtype=float)
Va   = np.array(Va,   dtype=float)
Vlp  = np.array(Vlp,  dtype=float)
Iμlp = np.array(Iμlp, dtype=float)
Iμls = np.array(Iμls, dtype=float)
iIμlp  = np.array(iIμlp,  dtype=float)
iIμls = np.array(iIμls, dtype=float)
```

In [267…
```
# Convert plate to volts
Vls = Vlpraw * 0.02
Vls = Vlsraw * 0.02

# Convert to current
Ilp = Vlp / 10_000
Ils = Vls / 10_000

# Convert to microamps
Iμlp = Ilp * 1000000
Iμls = Ils * 1000000

# Define per-run uncertainties
ΔV_meas = 0.01  # volts
ΔR_plate = 0.05 * 10_000

ΔIμlp = Iμlp * np.sqrt(((ΔV_meas / Vp1)**2) + ((ΔR_plate / 10_000)**2))
ΔIμls = Iμls * np.sqrt(((ΔV_meas / Vp2)**2) + ((ΔR_plate / 10_000)**2))

# Define extra fluctuation for points recorded as fluctuating extra
fluct_Iμlp = [10,23,26,27,39]
ΔIμlp[fluct_Iμlp] *= 2
# Some Values were fluctuating by 2 points
fluct_Iμlp = [11,25,36]
ΔIμlp[fluct_Iμlp] *= 3

fluct_Iμls = [8,23,24,26,27,28,29,40]
ΔIμls[fluct_Iμls] *= 2
# Value that fluctuated by 2
fluct_Iμls = [39]
ΔIμls[fluct_Iμls] *= 2
```

In [268…
```
#Save Values as Np arrays
ΔIμls=np.array(ΔIμls,dtype=float)
ΔIμlp=np.array(ΔIμlp,dtype=float)
```

4.1 Xenon vs Frozen plate currents compared

In [269…
```
plt.figure(figsize=(7,5))

# Main line
plt.plot(Va, Iμp, color='tab:red', linewidth=2, label='Xenon Plate Current')

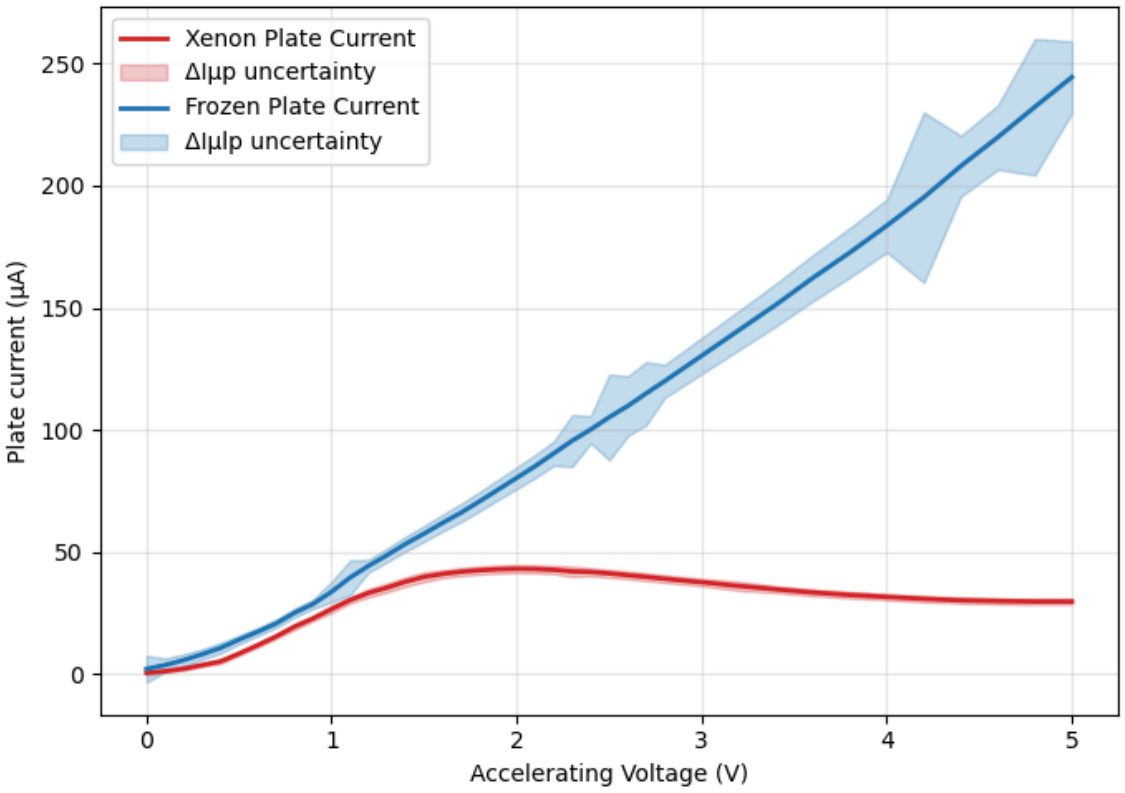
# Shaded uncertainty band
plt.fill_between(
    Va,
    Iμp - ΔIμp ,
```

```
    Iμp + ΔIμp,
    color='tab:red',
    alpha=0.25,
    label='ΔIμp uncertainty'
)

# Main line
plt.plot(Va, Iμlp, color='tab:blue', linewidth=2, label='Frozen Plate Current')

# Shaded uncertainty band
plt.fill_between(
    Va,
    Iμlp - ΔIμlp ,
    Iμlp + ΔIμlp,
    color='tab:blue',
    alpha=0.25,           # transparency of shading
    label='ΔIμlp uncertainty'
)

# Axis labels and styling
plt.xlabel("Accelerating Voltage (V)")
plt.ylabel("Plate current (μA)")
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



In [270…

```
plt.figure(figsize=(7,5))

# Main line
plt.plot(Va, Iμp, color='tab:red', linewidth=2, label='Xenon Plate Current')

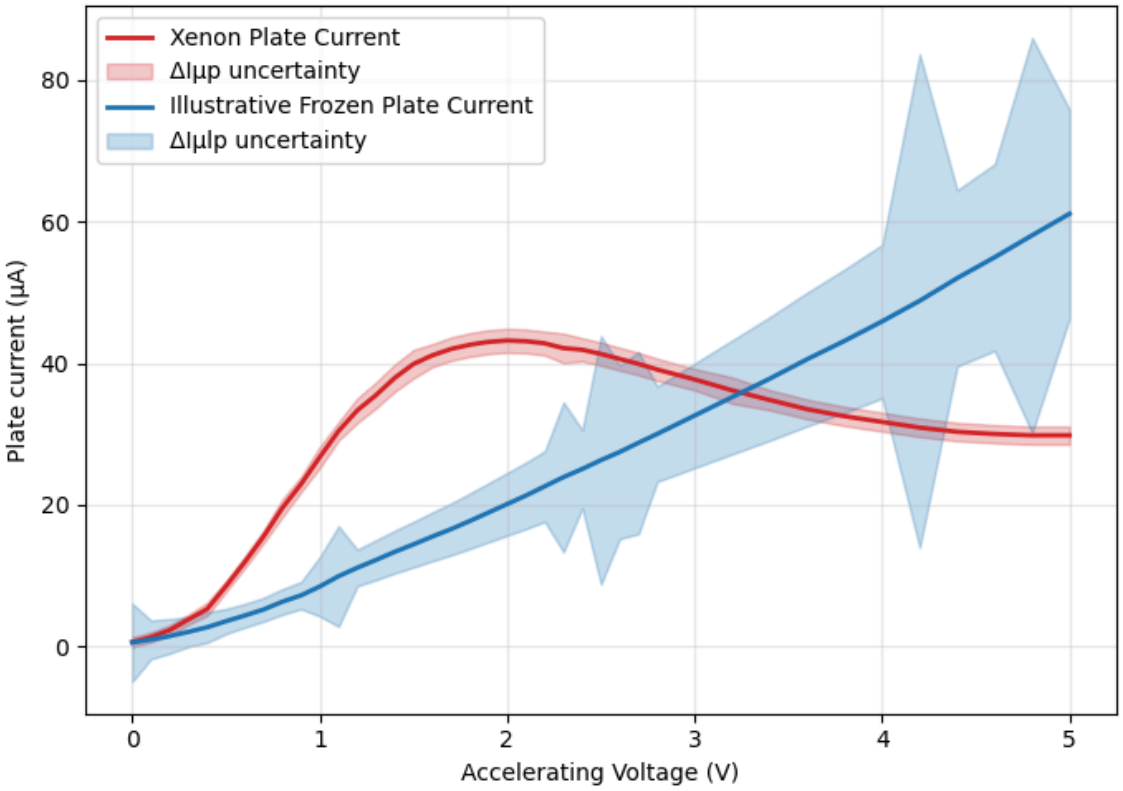
# Shaded uncertainty band
plt.fill_between(
```

```
    Va,
    Iμp - ΔIμp ,
    Iμp + ΔIμp,
    color='tab:red',
    alpha=0.25,          # transparency of shading
    label='ΔIμp uncertainty'
)

# Main line
plt.plot(Va, iIμlp, color='tab:blue', linewidth=2, label='Illustrative Frozen Pl

# Shaded uncertainty band
plt.fill_between(
    Va,
    iIμlp - ΔIμlp ,
    iIμlp + ΔIμlp,
    color='tab:blue',
    alpha=0.25,          # transparency of shading
    label='ΔIμlp uncertainty'
)

# Axis labels and styling
plt.xlabel("Accelerating Voltage (V)")
plt.ylabel("Plate current (μA)")
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



4.2 Probability of Scattering

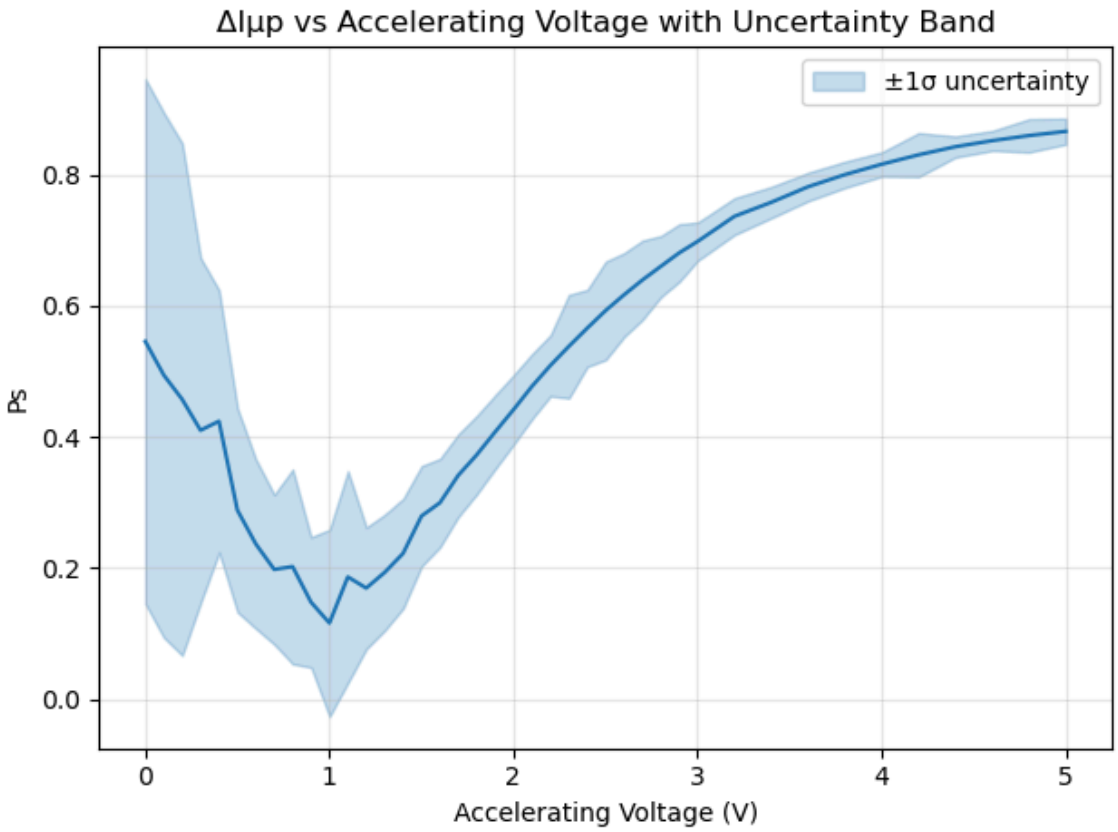
In [271…

```
import numpy as np
import matplotlib.pyplot as plt
```

```
# Compute Ps = 1 - (Iμls * Iμp) / (Iμs * Iμlp)
num = Iμls * Iμp
den = Iμs  * Iμlp
with np.errstate(divide='ignore', invalid='ignore'):
    ratio = np.where(den != 0, num / den, np.nan)
Ps = 1 - ratio

#  Compute propagated uncertainty ΔPs
with np.errstate(divide='ignore', invalid='ignore'):
    rel_err_sq = (
        (ΔIμls / np.where(Iμls != 0, Iμls, np.nan))**2 +
        (ΔIμp  / np.where(Iμp  != 0, Iμp,  np.nan))**2 +
        (ΔIμs  / np.where(Iμs  != 0, Iμs,  np.nan))**2 +
        (ΔIμlp / np.where(Iμlp != 0, Iμlp, np.nan))**2
    )
    ΔPs = np.abs(ratio) * np.sqrt(rel_err_sq)
#limit maximum uncertainty as it goes mad at low figures
ΔPs = np.clip(ΔPs, 0, 0.4)

plt.plot(Va,Ps)
#shade uncertainties
plt.fill_between(
    Va,
    Ps + ΔPs,
    Ps - ΔPs,
    color='tab:blue',
    alpha=0.25,
    label='±1σ uncertainty',
    )
plt.xlabel("Accelerating Voltage (V)")
plt.ylabel("Ps")
plt.title("ΔIμp vs Accelerating Voltage with Uncertainty Band")
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



4.2 Electron Momentum

In [272…

```
#Define Em
Em = np.sqrt(np.maximum(Va - Vs + 0.35, 0))

# y-uncertainty
dPs = np.clip(ΔPs, 0, 0.4)

# x-uncertainty from Vs ΔEm ≈ ΔVs / (2*Em)
ΔVs = Vs * np.sqrt((ΔV_meas / np.where(Vs != 0, Vs, np.nan))**2 + (ΔR_shield/100
dEm = (0.07+ΔVs / (2 * np.where(Em != 0, Em, np.nan)))
dEm = np.nan_to_num(dEm, nan=0.0, posinf=0.0, neginf=0.0)

# sort by Em so the shaded band draws cleanly
idx = np.argsort(Em)
Em, Ps, dPs, dEm = Em[idx], Ps[idx], dPs[idx], dEm[idx]

plt.plot(Em, Ps, color='tab:blue', label='Ps data')
plt.fill_between(Em, Ps - dPs, Ps + dPs, color='tab:blue', alpha=0.25, label='±1

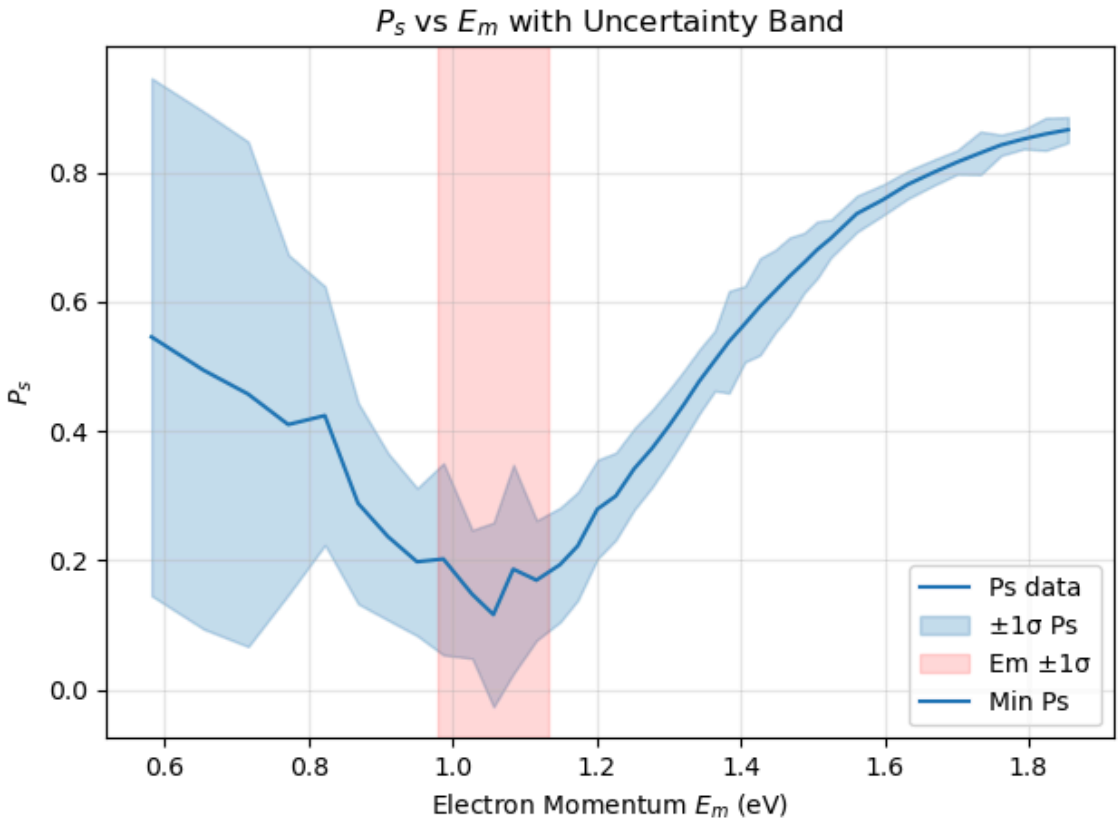
# minimum point
min_idx = np.nanargmin(Ps)
Em_min, Ps_min = Em[min_idx], Ps[min_idx]
dEm_min, dPs_min = dEm[min_idx], dPs[min_idx]

# show Em uncertainty
plt.axvspan(Em_min - dEm_min, Em_min + dEm_min, color='red', alpha=0.15, label='
plt.plot([Em_min], [Ps_min], label='Min Ps')

# labels
plt.xlabel("Electron Momentum $E_m$ (eV)")
```

```
plt.ylabel("$P_s$")
plt.title("$P_s$ vs $E_m$ with Uncertainty Band")
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

print(f"Minimum Ps = {Ps_min:.4f} ± {dPs_min:.4f} at Em = {Em_min:.4f} ± {dEm_mi
```




```
Minimum Ps = 0.1164 ± 0.1424 at Em = 1.0564 ± 0.0773 eV
```

In [273…

```
# numpy array
Ps=np.array(Ps, dtype=float)
```

4.3 Reciprocal Mean Free path

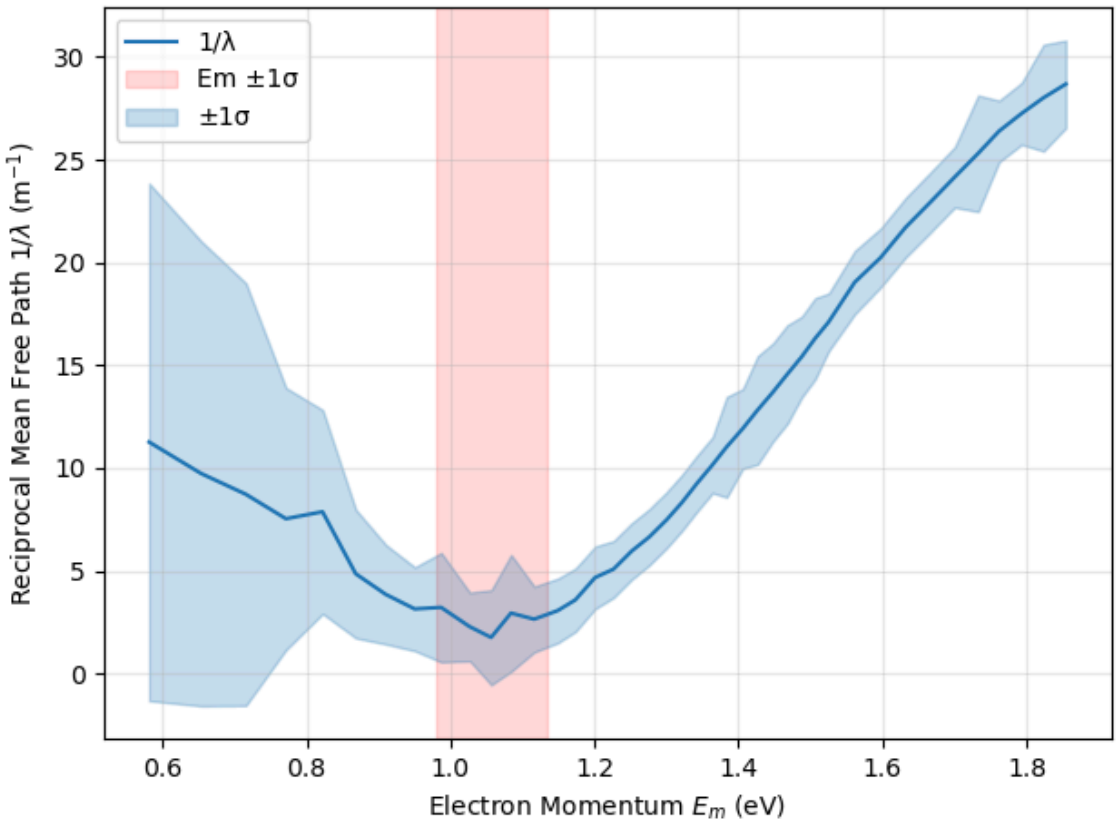
In [274…

```
# deine l = 0.07 m and define invfmp from eq 11
l = 0.07
fmp = -l / np.log(1 - Ps)
invfmp = 1 / np.clip(fmp, 1e-12, None)
# uncertainity definition propogated with Ps uncertainity
Δl = 0.001
Δfmp = np.abs(fmp) * np.sqrt(
    (Δl / l)**2 + (ΔPs / ((1 - Ps) * np.log(np.clip(1 - Ps, 1e-12, None))))**2
)

Δinvfmp = Δfmp / (fmp**2)
Δfmp = np.nan_to_num(Δfmp, nan=0.0, posinf=0.0, neginf=0.0)
Δinvfmp = np.nan_to_num(Δinvfmp, nan=0.0, posinf=0.0, neginf=0.0)
# Find min point- same as Ps
min_idx = np.nanargmin(invfmp)
Em_min, invfmp_min = Em[min_idx], invfmp[min_idx]
dEm_min, Δinvfmp_min = dEm[min_idx], Δinvfmp[min_idx]
```

```
# Plot
plt.plot(Em, invfmp, color='tab:blue', label='1/λ')
plt.axvspan(Em_min - dEm_min, Em_min + dEm_min, color='red', alpha=0.15, label='
plt.fill_between(Em, invfmp - Δinvfmp, invfmp + Δinvfmp,
                 color='tab:blue', alpha=0.25, label='±1σ')
plt.xlabel("Electron Momentum $E_m$ (eV)")
plt.ylabel(r"Reciprocal Mean Free Path $1/\lambda$ (m$^{-1}$)")
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

print(f"Minimum inv free path = {invfmp_min:.4f} ± {Δinvfmp_min:.4f} at Em = {Em
```




```
Minimum inv free path = 1.7672 ± 2.3022 at Em = 1.0564 ± 0.0773 eV
```

4.4 Probability of Collision

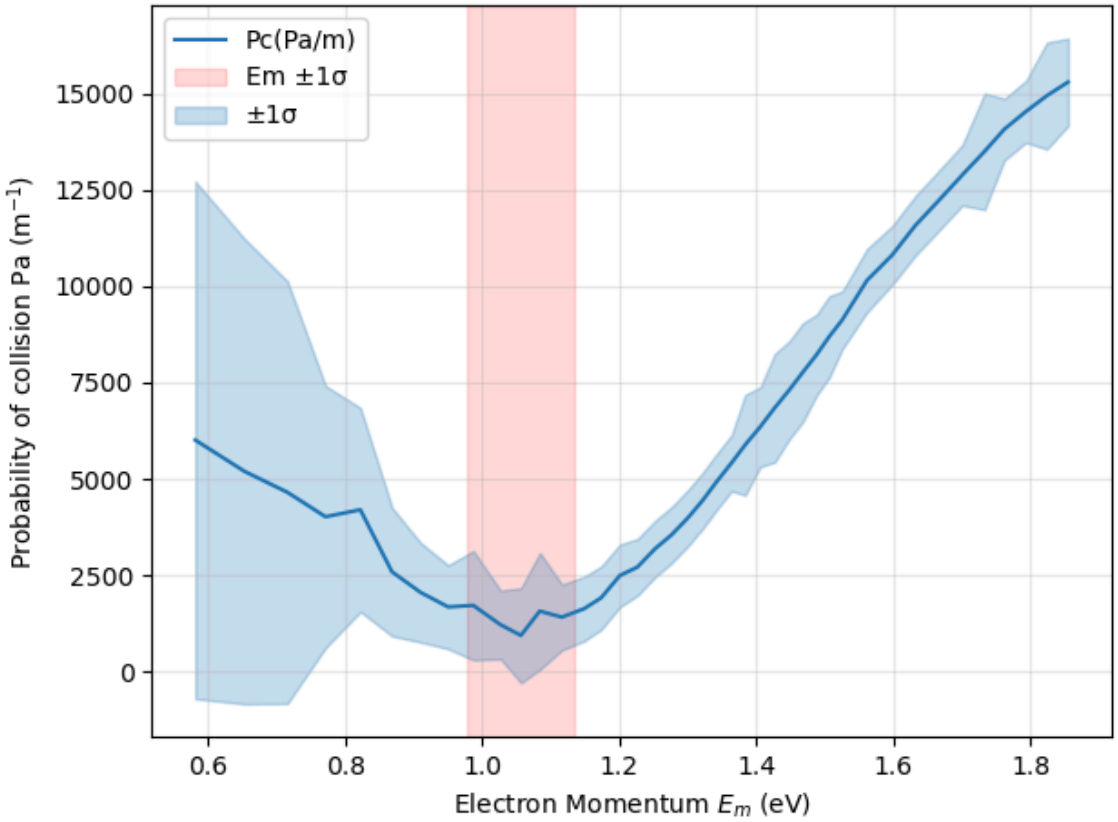
In [275…

```
# Multiply by 4(133.3) to convert from Torr to Pa and multiply by this new press
Pc= 4*invfmp*133.3
ΔPc= 4*Δinvfmp*133.3
#Min point
min_idx = np.nanargmin(Pc)
Em_min, Pc_min = Em[min_idx], Pc[min_idx]
dEm_min, ΔPc_min = dEm[min_idx], ΔPc[min_idx]

plt.plot(Em, Pc, color='tab:blue', label= 'Pc(Pa/m)')
plt.axvspan(Em_min - dEm_min, Em_min + dEm_min, color='red', alpha=0.15, label='
plt.fill_between(Em, Pc - ΔPc, Pc + ΔPc,
                 color='tab:blue', alpha=0.25, label='±1σ')
#Plot
plt.xlabel("Electron Momentum $E_m$ (eV)")
plt.ylabel(r"Probability of collision Pa (m$^{-1}$)")
```

```
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

print(f"Minimum Pc = {Pc_min:.4f} ± {Pc_min:.4f} at Em = {Em_min:.4f} ± {dEm_min
```




```
Minimum Pc = 942.2887 ± 942.2887 at Em = 1.0564 ± 0.0773 eV
```

4.5 Corrected Electron momentum for Probability of Colission

In [276…

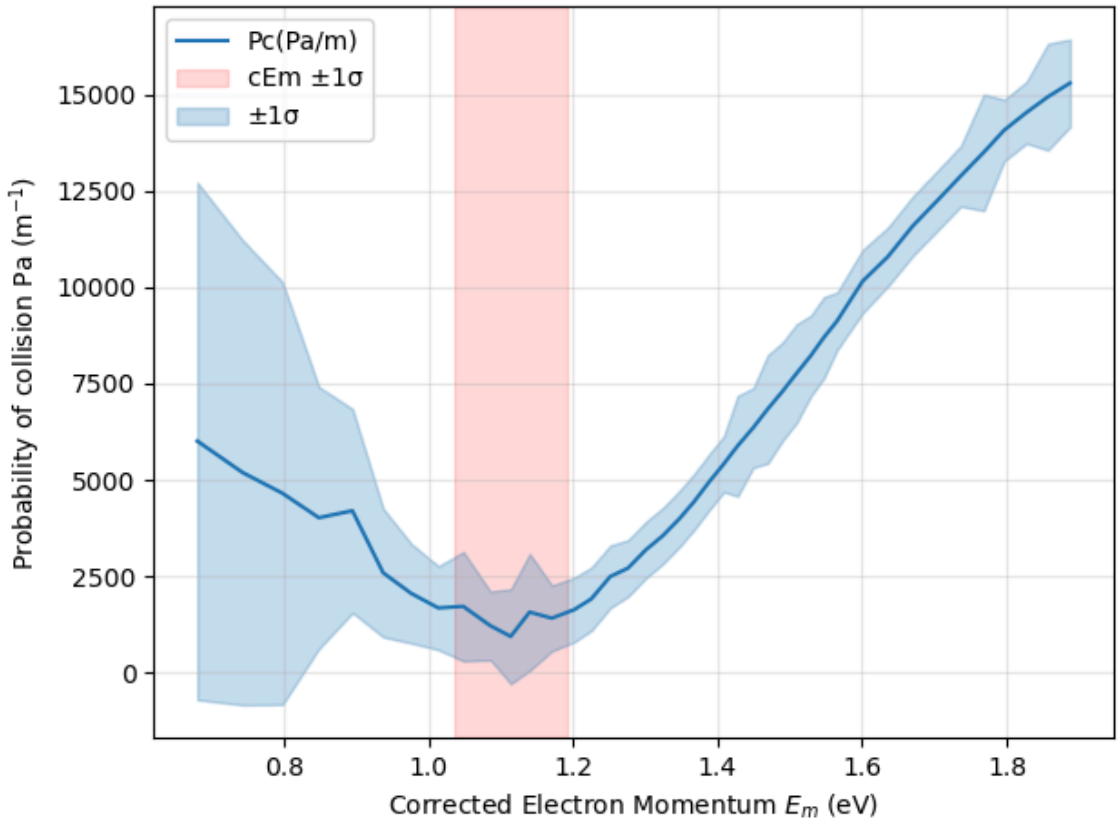
```
cEm = np.sqrt(np.maximum(Va - Vs + 0.262 + 0.212, 0))
cdEm = ((0.018 + 0.048)+ΔVs / (2 * np.where(cEm != 0, cEm, np.nan)))
cdEm = np.nan_to_num(dEm, nan=0.0, posinf=0.0, neginf=0.0)

# Same code as previous plot copy paste with variable name change and sub in val

min_idx = np.nanargmin(Pc)
cEm_min, Pc_min = cEm[min_idx], Pc[min_idx]
cdEm_min, ΔPc_min = cdEm[min_idx], ΔPc[min_idx]

plt.plot(cEm, Pc, color='tab:blue', label= 'Pc(Pa/m)')
plt.axvspan(cEm_min - cdEm_min, cEm_min + cdEm_min, color='red', alpha=0.15, lab
plt.fill_between(cEm, Pc - ΔPc, Pc + ΔPc,
                 color='tab:blue', alpha=0.25, label='±1σ')
plt.xlabel("Corrected Electron Momentum $E_m$ (eV)")
plt.ylabel(r"Probability of collision Pa (m$^{-1}$)")
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```

```
print(f"Minimum Pc = {Pc_min:.4f} ± {Pc_min:.4f} at cEm = {cEm_min:.4f} ± {cdEm_
```




```
Minimum Pc = 942.2887 ± 942.2887 at cEm = 1.1136 ± 0.0773 eV
```

4.5 Extension Choosing contact potential based on Graph

In [321…

```
#Visually chosen turning point as graph was here and has disappeared. Code was s
x0 = -0.27

# Find the data point closest to that x0
i = np.argmin(np.abs(Vsee - x0))
#store points of associated variables
x_turn = Vstee[i]
y_turn = lnse[i]
x_unc = Vsteeunc[i]
y_unc = lnsteunc[i]
```

In [322…

```
# Percent uncertainties
px = abs(Vsteeunc[i] / x)
py = abs(lnsteunc[i])

# Combine percentages
p_quad = (px**2 + lnsteunc[i]**2)**0.5

# Convert to absolute uncertainty in x
absunc = abs(x) * p_quad
#Print results
print(x,absunc)
```

```
-0.262 0.04807055000494355
```

4.5 Now Contact potential has been established and found plot Contact potential visually (Data Store)

In [323…

```
Vae = np.array([0, 0.05, 0.10, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4, 0.45])
Vsrawe1 = np.array([1.0, 0.7, 0.5, 0.4, 0.3, 0.2, 0.1, 0.1, 0.1, 0])
Vsrawe2 = np.array([0.9, 0.8, 0.6, 0.4, 0.3, 0.2, 0.1, 0.1, 0.0, 0 ])
Vsrawe3 = np.array([0.9, 0.7, 0.5, 0.4, 0.3, 0.2, 0.1, 0, 0 ,0])

# Average the three runs
Vsrawe = (Vsrawe1 + Vsrawe2 + Vsrawe3)

# Fixed uncertainties  ,±0.1 V on shield readings, ±0.02 V on accelerating volta
Vsraweunc = 0.1 * np.ones_like(Vsrawe)
Vaeunc = 0.02 * np.ones_like(Vae)

# Convert to shield voltage and uncertainty
Vste = Vsrawe * 0.02
Vsteunc = Vsraweunc * 0.02

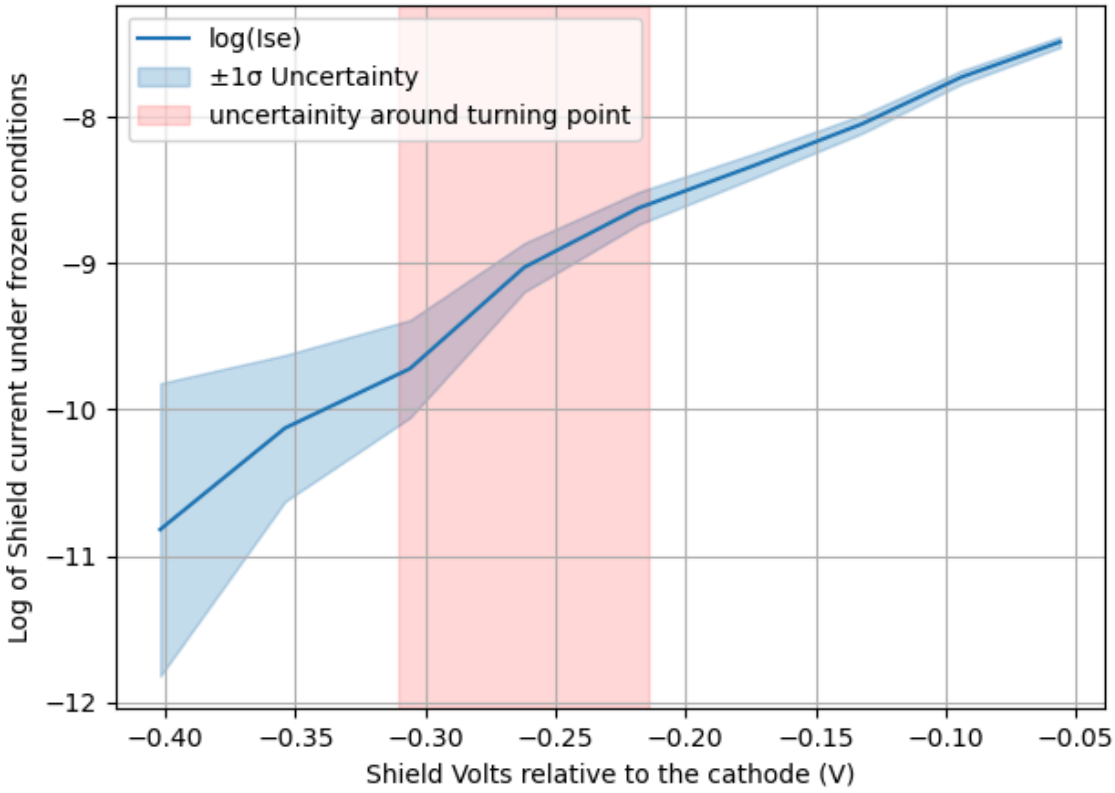
# Convert to current (A), Rs = 100 Ω
Iste = Vste / 100
Isteunc = Vsteunc / 100
#uncertianty of 0.5 ohms

# Mask zero or negative values (to avoid log errors)
mask = Iste > 0
Iste = Iste[mask]
Isteunc = Isteunc[mask]
Vae = Vae[mask]
Vaeunc = Vaeunc[mask]
Vste = Vste[mask]
Vsteunc = Vsteunc[mask]

# Natural log of current and propagated uncertainty
lnste = np.log(Iste)
lnsteunc = Isteunc / Iste

# Effective accelerating voltage: Vsee = -(Vse + Vae)
Vstee = -(Vste + Vae)
# Propagate uncertainty in Vsee
Vsteeunc = np.sqrt(Vsteunc**2 + Vaeunc**2)

plt.figure(figsize=(7,5))
plt.plot(Vstee, lnste, color='tab:blue', label='log(Ise)')
plt.fill_between(Vstee,
                 lnste - lnsteunc,
                 lnste + lnsteunc,
                 color='tab:blue',
                 alpha=0.25,
                 label='±1σ Uncertainty')
plt.axvspan(x - absunc,x + absunc, color='red', alpha=0.15, label='uncertainity
plt.xlabel('Shield Volts relative to the cathode (V)')
plt.ylabel('Log of Shield current under frozen conditions')
plt.grid(True)
plt.legend()
plt.show()
```



4.5 Thermionic Emission

In [324…

```
Vprawe1= np.array([1.0,0.7,0.6,0.4,0.2,0.1,0,0,0])
Vprawe2= np.array([0.9,0.7,0.5,0.4,0.3,0.2,0.1,0,0])
Vprawe3= np.array([1.0,0.8,0.5,0.3,0.2,0.1,0,0,0])

Vprawe=np.array(((Vprawe1+Vprawe2+Vprawe3)/3)*(0.02))
Vpunc=0.1
```

In [287…

```
plt.plot(Vprawe, lnste, color='tab:blue', label='log(Ise)')
plt.fill_between(Vprawe,
                 lnste - lnsteunc,
                 lnste + lnsteunc,
                 color='tab:blue',
                 alpha=0.25,
                 label='±1σ Uncertainty')
# Choose linear section of graph to take slope of
mask = (Vprawe >= 0.0030) & (Vprawe <= 0.125)
w = 1.0 / np.clip(lnsteunc[mask], 1e-12, None)
(coefs, cov) = np.polyfit(Vprawe[mask], lnste[mask], 1, w=w, cov=True)
m, c = coefs
m_unc = float(np.sqrt(cov[0, 0]))  # 1σ on slope
c_unc = float(np.sqrt(cov[1, 1]))  # 1σ on interce

fit_line = np.poly1d([m, c])

x_fit = np.linspace(Vprawe[mask].min(), Vprawe[mask].max(), 100)
y_fit = fit_line(x_fit)

# Plot the best-fit line and print its equation
plt.plot(x_fit, y_fit, 'r--', label='Best-fit line')
print("Slope:", round(m, 4))
```

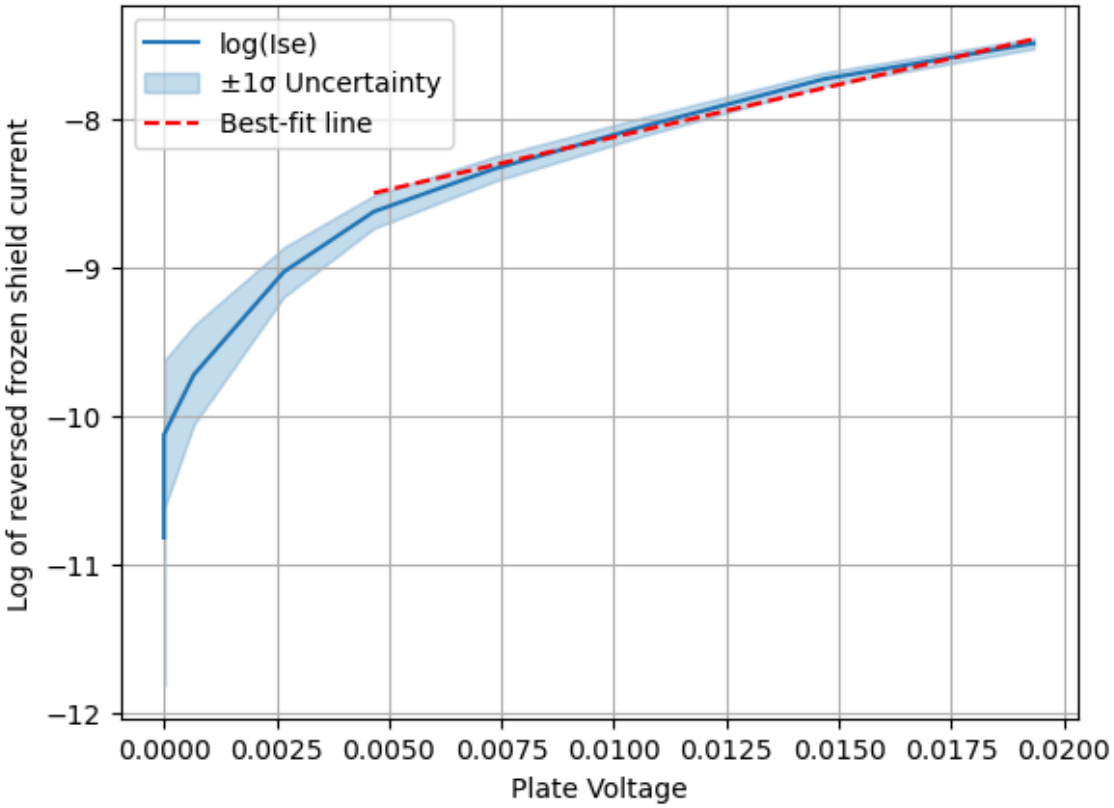
```
print("Intercept:", round(c, 4))


# Finish plot (you likely already have this)
plt.xlabel('Plate Voltage')
plt.ylabel('Log of reversed frozen shield current')
plt.grid(True)
plt.legend()
plt.show()

# average thermionic electron energy in volts
Vbar = -3.0 / (2.0*m)

print("Slope m =", m)
print("Intercept c =", c)
print("Average thermionic energy (Vbar) =", Vbar, "V")
```

```
Slope: 70.7661
Intercept: -8.8275
```




```
Slope m = 70.76606761449122
Intercept c = -8.827548060924519
Average thermionic energy (Vbar) = -0.021196599593063095 V
```